Review of “Synthetic Difference-in-Differences”*

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1 Summary

This paper proposes a novel method of causal inference using panel data—synthetic difference-in-differences (SDID). The estimation takes advantage of the strengths of traditional difference-in-differences (DID) and synthetic control methods (SCM) while mitigating their respective weaknesses. The contextual contribution is therefore twofold: building on the long line of established research in DID—dating back to the likes of Ashenfelter and Card (1985); Card and Krueger (1994)—and providing an important innovation in the more contemporary domain of SCM, which saw its inception in the works of Abadie and Gardeazabal (2003) and Abadie et al. (2010;2015). The authors' contribution also extends theoretically, where they establish asymptotic normality of the SDID estimator. Finally, Arkhangelsky et al. (2021) validate their method empirically by applying SDID to the 1988 California tobacco cessation programme (Abadie et al. 2010) and conducting simulation studies calibrated to (1) the Current Population Survey (CPS) as in Bertrand et al. (2004), and (2) the Penn World Table (Feenstra et al. 2015).

The primary innovation of the paper is the following SDID estimator:[1]

$$\left(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}\right) = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \hat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} \right\} \tag{1}$$

Similar to SCM, the method reweights and matches on pre-treatment trends, eliminating traditional DID's strong reliance on the parallel trends assumption. Furthermore, the estimator uses time weights to emphasise pre-treatment periods that best predict post-treatment outcomes for control units. These weights are captured by $\hat{\omega}_i^{sdid}$ and $\hat{\lambda}_i^{sdid}$, respectively. Similar to DID, the method is invariant to additive unit-level shifts allowing for valid large-panel inference, eliminating SCM's core weakness. Thereby, Arkhangelsky et al. (2021) argues that (1) achieves double-robustness while eliminating bias.

*Arkhangelsky, Athey, Hirshberg, Imbens, and Wager (2021).

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[1] $\hat{\tau}^{sdid}$ estimates the average treatment effect (ATT) per post-treatment period. Arkhangelsky et al. (2021) elaborate on this interpretation and provide direct comparisons with DID and SCM in Section I.B.

A secondary innovation relates to the optimisation problem assigning unit weights $\hat{\omega}_i^{sdid}$:

$$(\hat{\omega}_0, \hat{\omega}^{sdid}) = \underset{\omega_0 \in \mathbb{R}, \omega \in \Omega}{\arg\min} \; \ell_{unit}(\omega_0, \omega), \tag{2}$$

where

$$\ell_{unit}(\omega_0, \omega) = \sum_{t=1}^{T_{pre}} \left(\omega_0 + \sum_{i=1}^{N_{co}} \omega_i Y_{it} - \frac{1}{N_{tr}} \sum_{i=N_{co}+1}^{N} Y_{it} \right)^2 + \zeta^2 T_{pre} \|\omega\|_2^2,$$

and Ω represents the standard SCM constraints to unit weights.[2] Unlike standard SCM, Arkhangelsky et al. (2021) include an intercept term ω_0 when estimating unit weights. This allows $\hat{\omega}^{sdid}$ to match parallel pre-trends without requiring exact level matching, since any constant unit differences are absorbed by the fixed effects α_i in (1). Furthermore, the authors include a penalty term to ensure uniqueness of the weights, as in Doudchenko and Imbens (2016). The time weights $\hat{\lambda}^{sdid}$ are another key departure from SCM, obtained by solving:

$$(\hat{\lambda}_0, \hat{\lambda}^{sdid}) = \underset{\lambda_0 \in \mathbb{R}, \lambda \in \Lambda}{\arg\min} \; \ell_{time}(\lambda_0, \lambda), \tag{3}$$

where

$$\ell_{time}(\lambda_0, \lambda) = \sum_{i=1}^{N_{co}} \left(\lambda_0 + \sum_{t=1}^{T_{pre}} \lambda_t Y_{it} - \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it} \right)^2,$$

and Λ is analogous to the Ω constraints for the temporal case. This temporal localisation improves precision by emphasising pre-periods with the best predictive power of post-treatment outcomes. It is important to note that the regularisation penalty does not appear in ℓ_{time}, reflecting the authors' assumption that observations may be correlated within units over time, but not across units within the same time period.

While I will skip over the formal results (Section III in Arkhangelsky et al. 2021), there is a powerful property of (1) that demands attention: double-robustness. That is, the estimator remains consistent if *either* the unit weights successfully balance pre-trends *or* the time weights successfully predict post-treatment outcomes for controls. Additionally, the use of two-way fixed effects in (1) and intercept terms ω_0 and λ_0 in (2) and (3) makes the SDID estimator invariant to additive shocks to any unit or time.

2 Evaluation

Arkhangelsky et al. (2021) conclude that SDID is a robust, flexible estimator that outperforms both DID and SCM, offering reliable causal inference with a sufficiently large sample. They further corroborate this finding by (1) establishing asymptotic normality under weaker assumptions than comparable alternatives, (2) providing empirical evidence via replication of the California smoking cessation programme (Abadie et al. 2010), and (3) illustrating

[2] I.e., $\omega_i \geq 0$ and $\sum_{i=1}^{N_{co}} w_i = 1$.

that SDID shows lower bias and RMSE across diverse data-generating processes in the CPS and Penn World Table placebo simulations. Indeed, the proposed SDID method has several attractive features, including intuitive interpretation of causal estimates. However, I see several drawbacks to using this method.

The primary concern is shared with synthetic control: unreliable inference. Trustworthy standard errors are difficult to obtain when the number of treated units is small—such as $N_{tr} = 1$, as in the California smoking example. In these cases, the only feasible variance estimator is the placebo method which relies on the strong assumption of homoskedasticity across treated and control units. Alternatively, there is the closely related permutation tests, which requires an equally strong assumption of exchangeability. Nevertheless, when working in a comparative case study setting with carefully selected control groups, these assumptions may be more plausible. Furthermore, while formal inference is complicated, SDID point estimates are arguably superior to DID and SCM alternatives, and for some applied research questions, a credible point estimate may be sufficient even without precise uncertainty quantification.

Another point of concern relates to the handling of covariates. While Arkhangelsky et al. (2021) note that SDID can be extended to account for covariates by applying Algorithm 1 to the residuals

$$Y_{it}^{res} = Y_{it} - X_{it}\hat{\beta} \tag{4}$$

where $\hat{\beta}$ is estimated via a preliminary regression of Y_{it} on X_{it}, they provide little guidance on how this is done. First, the form of this regression is unclear: two-way fixed effects or a simple restrictive regression? Should it be estimated on the full sample or only controls? Second, the use of covariates is missing from both their empirical application and simulation studies, which leaves one to question how it effects SDID's performance in practice. Third, although the R-package `synthdid` provides the option to include covariates in the main estimation function via a three-dimensional array of time-varying covariates $(N \times T \times C)$, there is little guidance as to how this 3D array is actually incorporated, leaving the applied economist to compute residuals Y_{it}^{res} themselves.[3]

Finally, recall that in Abadie et al. (2010), the predictor weight matrix $\mathbf{V}$ is chosen via data-driven methods to minimise pre-treatment prediction error, allowing covariates that strongly predict the outcome better to receive more weight in the matching. SDID's residualisation approach loses this nuanced weighting. While this may be a design choice, I believe it is of sufficient importance that Arkhangelsky et al. (2021) should have mentioned this departure from SCM when describing the key differences between their proposed SDID and the original SCM.

In the next section, we will replicate Andersson (2019)'s analysis of Sweden's carbon tax to empirically evaluate these concerns and directly compare estimates from SDID, DID, and SCM.

[3]As of writing, there are multiple issues open on the package's Github page on this topic. I could not get functioning covariates using the provided function during my replication exercise, either. All else considered, it is a fantastic statistical package.

3 Replication: Andersson (2019)

In order to test the novel estimation method, I apply SDID to the case of a 1990–1991 revision of the Swedish carbon tax regime. The change was comprised of (1) the extension of Sweden's existing value-added-tax (VAT) of 25 percent on the retail price of gasoline and diesel in 1990, and (2) the implementation of the Swedish carbon tax of US$30 per ton of CO2 in the following year. Table 1 shows the cleaned panel data sample consisting of 14 OECD countries as provided in the online replication package of Andersson (2019). The identification assumptions are sufficiently discussed in Section II.A of their paper and are, therefore, omitted here.

Table 1: Data Used in Andersson (2019)

Category	Variable	Measurement	Source
Outcome	CO_2 emissions from transport	Metric tons per capita	World Bank (2015)
Predictor	GDP per capita (PPP)	2005 USD	Feenstra et al. (2015)
	Motor vehicles per 1000 people		Dargay et al. (2007)
	Gasoline consumption per capita	kg oil equivalent	World Bank (2015)
	Urban population	% of total	World Bank (2015)
	Lagged CO_2 emissions	Metric tons per capita (1970, 1980, 1989)	World Bank (2015)

Notes: Predictors are averaged over the 1980–1989 period. The synthetic control is constructed using a donor pool of 14 OECD countries (Australia, Belgium, Canada, Denmark, France, Greece, Iceland, Japan, New Zealand, Poland, Portugal, Spain, Switzerland, USA) that did not implement carbon taxes covering transport during the sample period (1960–2005).

As a starting point, I test the SDID procedure on the data excluding covariates. Figure 1 reports the results in comparison to DID and SCM. Similar to the results from Arkhangelsky et al. (2021)'s empirical application on the California smoking study, we find that SDID is effective in balancing weights more evenly by punishing overweighting and major dispersion. Interestingly, 1989 receives nearly the entire time weight (0.926) while the remainder is attributed to 1962. Furthermore, as shown in Table 2, SDID's point estimate is substantially larger than DID or SCM for the case without covariates. This begs the question: To what extent is this difference attributable to SDID's temporal localisation through time weighting, particularly the concentration on the immediate pre-treatment year? Nevertheless, similar to Andersson (2019)'s results (minding that their SCM estimates include covariate weighting), the SDID estimator assigns the largest weights to Denmark (0.164), United States (0.127), Belgium (0.104), New Zealand (0.103) and Swizterland (0.100). Together, these countries make up roughly 60 percent of synthetic Sweden in SDID specification (2). Intuitively, these weights make sense since Denmark is economically and culturally the closest comparison to Sweden, Belgium and Sweden have enjoyed similar GDP growth rates, and New Zealand and Sweden have comparable population densities and geographies.

Andersson (2019) makes a good case for the use of observables in his identifying assumptions; I therefore extend my replication to account for exogenous time-varying covariates.

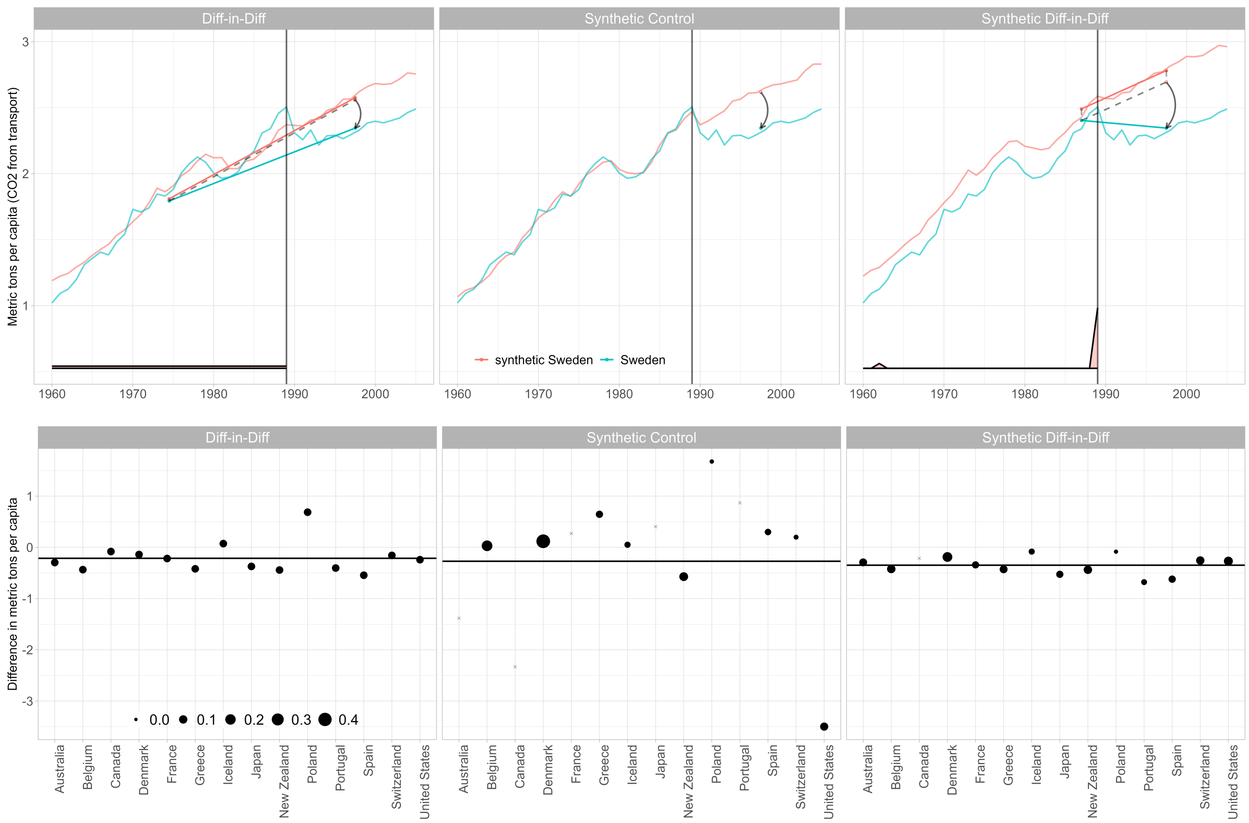

Figure 1: A comparison of DID, SC, and SDID estimates without covariates.

Notes: Top panel: Trends in CO_2 emissions over time for Sweden and the weighted average of its OECD control counterparts. Time weights used to average pretreatment periods are reported at the bottom. Bottom panel: Country-by-size adjusted outcome differences (as discussed in Section I.B of Arkhangelsky et al., 2021). Weights are indicated by dot size; the estimated effect is shown by the horizontal line. Countries receiving zero weights are indicated by an × symbol.

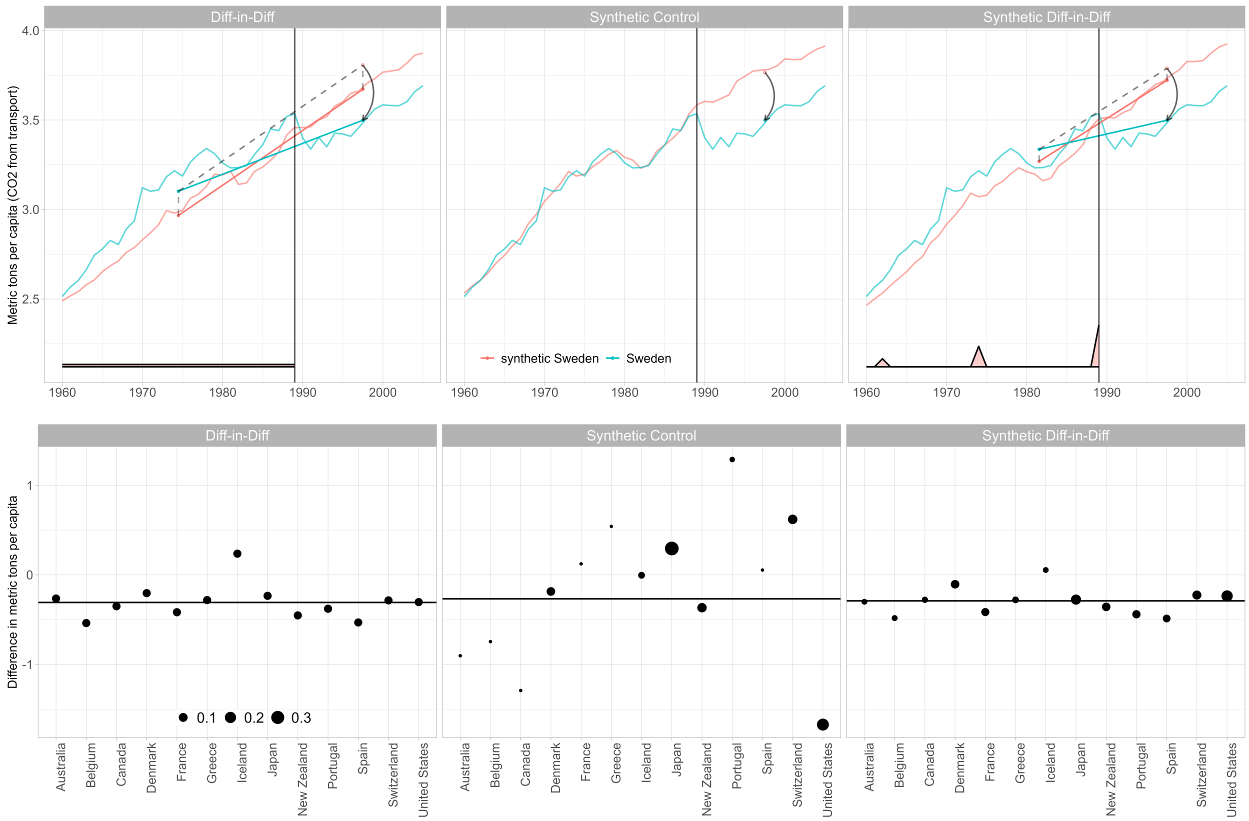

Figure 2: A comparison of DID, SC, and SDID estimates on covariate adjusted outcomes, as proposed by Kranz (2022).

Recall that Arkhangelsky et al. (2021) propose accounting for covariates by applying Algorithm 1 to the residuals as in Equation (4). Clarke et al. (2023) point out that accounting for covariates in this manner may be sensitive to scaling, especially when these observables have large magnitudes and variances. Kranz (2022) further contends that bias may be introduced if covariates are correlated with unobserved unit or time effects. They propose instead estimating $\hat{\beta}$ via a simple two-way fixed effects regression with time and unit fixed effects on untreated observations

$$Y_{it} = \alpha_i + \mu_t + X_{it}\beta + \varepsilon_{it}, \quad \text{for } W_{it} = 0,$$

and then constructing adjusted outcomes $Y_i t^{adj}$ using the method in Equation (4).

Since `synthdid` requires balanced panels, we drop Poland from our data due to missing GDP data for 1960–1969. Furthermore, Kranz (2022)'s `xsynthdid` package automatically drops lagged CO2 emissions due to collinearity. Hence, only four predictors are used. Specification (3) of Table 2 and Figure 2 report the results for this method. The inclusion of covariates significantly alters the temporal and spatial weights. Compared to no-covariates specification (2), time weights in specification (3) are now more balanced between 1989 (0.595), 1974 (0.291), and 1962 (0.114). This suggests that covariates may help SDID identify a longer term counterfactual trend, rather than placing all the focus on the immediate pre-treatment period. Spatially, the donor pool weights have shifted to favour the United States (0.204), Japan (0.150), Switzerland (0.106), Denmark (0.090), and New Zealand (0.089) the most. These changes demonstrate that covariate choice substantively shapes the counterfactual construction. Furthermore, more dispersed time weights may produce estimates that are less sensitive to pre-treatment shocks in any single year.

Table 2: Comparison of Treatment Effect Estimates: DID, SCM, and SDID

	(1) Andersson (2019)	(2) Arkhangelsky et al. (2021)	(3) Kranz (2022)
DID	-0.214	-0.214	-0.308
	(0.085)	(0.313)	(0.185)
SCM	-0.290	-0.271	-0.267
		(0.422)	(0.300)
SDID		-0.348	-0.290
		(0.326)	(0.233)

Notes: Standard errors are shown in parentheses. **(1)** Andersson (2019) specification: DID estimates are found using a TWFE regression without covariates. Standard errors reported are clustered by country. **(2)** Arkhangelsky et al. (2021) specification without covariates: SEs are calculated by placebo method. **(3)**: Incorporates covariates by running SDID on the TWFE residualised outcome Y_{it}^{adj}.

To draw a direct comparison with Andersson (2019)'s results, it would be ideal to incorporate covariate weighting as in traditional SCM (e.g., Abadie et al. 2010), where predictor importance is explicitly weighted via a $\mathbf{V}$ matrix. Developing such estimator, however, lies beyond the scope of this review. I should note, however, that there some preliminary research on integrating covariates into the SDID framework in a similar manner (see Hirshberg and Klosin 2024).

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program. *Journal of the American Statistical Association*, 105(490):493–505.

Abadie, A., Diamond, A., and Hainmueller, J. (2015). Comparative Politics and the Synthetic Control Method. *American Journal of Political Science*, 59(2):495–510.

Abadie, A. and Gardeazabal, J. (2003). The Economic Costs of Conflict: A Case Study of the Basque Country. *American Economic Review*, 93(1):113–132.

Andersson, J. J. (2019). Carbon Taxes and CO2 Emissions: Sweden as a Case Study. *American Economic Journal: Economic Policy*, 11(4):1–30.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic Difference-in-Differences. *American Economic Review*, 111(12):4088–4118.

Ashenfelter, O. and Card, D. (1985). Using the Longitudinal Structure of Earnings to Estimate the Effect of Training Programs. *The Review of Economics and Statistics*, 67(4):648–660.

Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How Much Should We Trust Differences-in-Differences Estimates? *The Quarterly Journal of Economics*, 119(1):249–275.

Card, D. and Krueger, A. B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*, 84(4):772–793.

Clarke, D., Pailañir, D., Athey, S., and Imbens, G. (2023). Synthetic Difference In Differences Estimation.

Dargay, J., Gately, D., and Sommer, M. (2007). Vehicle Ownership and Income Growth, Worldwide: 1960-2030. *The Energy Journal*, Volume 28(Number 4):143–170.

Doudchenko, N. and Imbens, G. W. (2016). Balancing, Regression, Difference-In-Differences and Synthetic Control Methods: A Synthesis. NBER Working Papers 22791, National Bureau of Economic Research, Inc.

Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The Next Generation of the Penn World Table. *American Economic Review*, 105(10):3150–82.

Hirshberg, D. and Klosin, S. (2024). Synthetic Differences-in-Differences with Covariates. Working paper. Draft: `SDIDC-WP.pdf`.

Kranz, S. (2022). Synthetic Difference-in-Differences with Time-Varying Covariates. Technical report. Available online at: `https://github.com/skranz/xsynthdid/blob/main/paper/synthdid_with_covariates.pdf`.

World Bank (2015). World Development Indicators. World Bank. Data retrieved from World Development Indicators database.